

GRID PULSE CHALLENGE

REAL-TIME DECISION SUPPORT FOR TRANSMISSION GRIDS

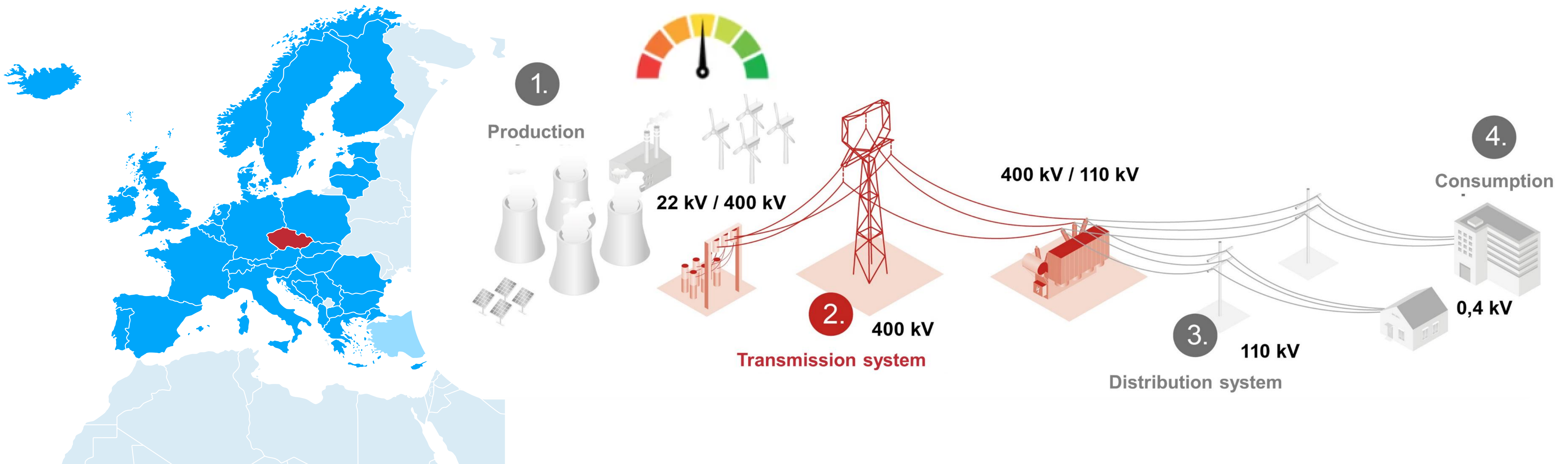
Enhancing grid management through advanced real-time data tools

GreenHack 2026



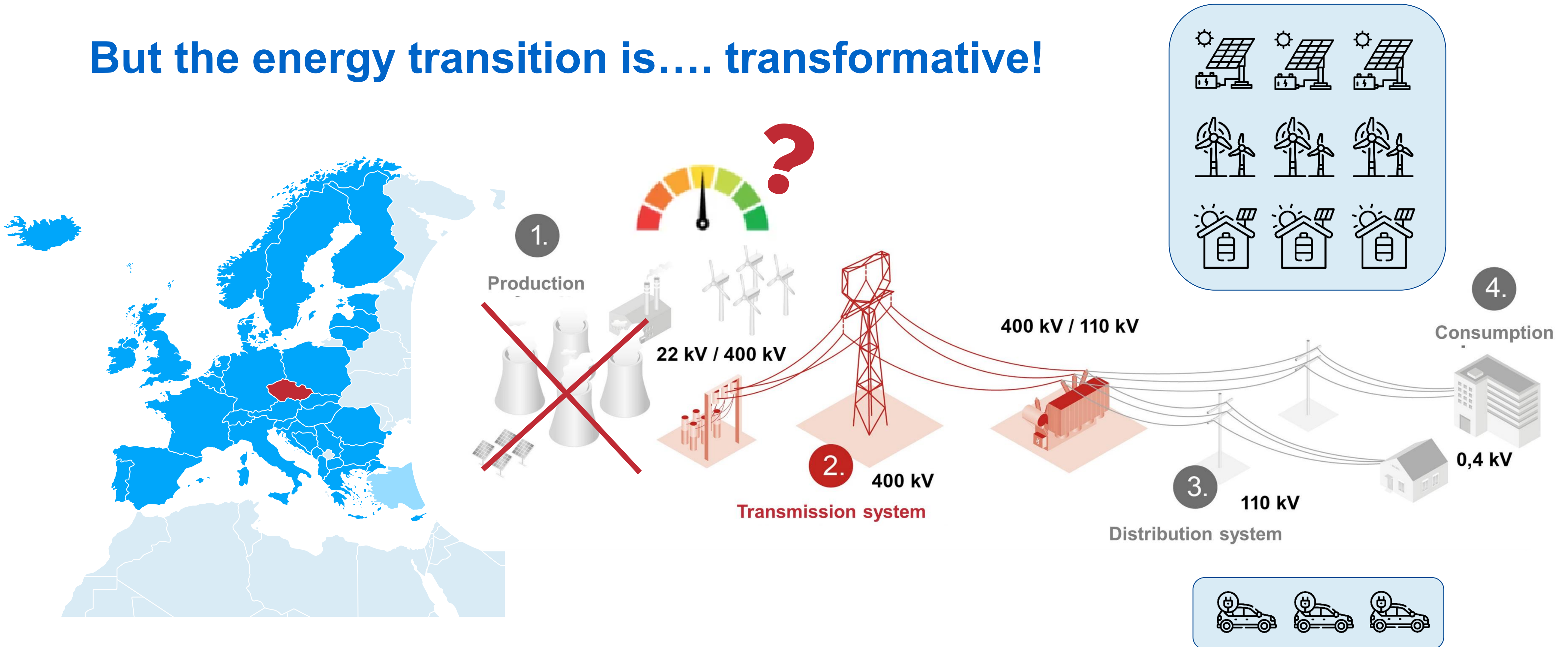
INTRODUCING ČEPS, a.s.

ČEPS is a TSO – Transmission System Operator for ELECTRICITY



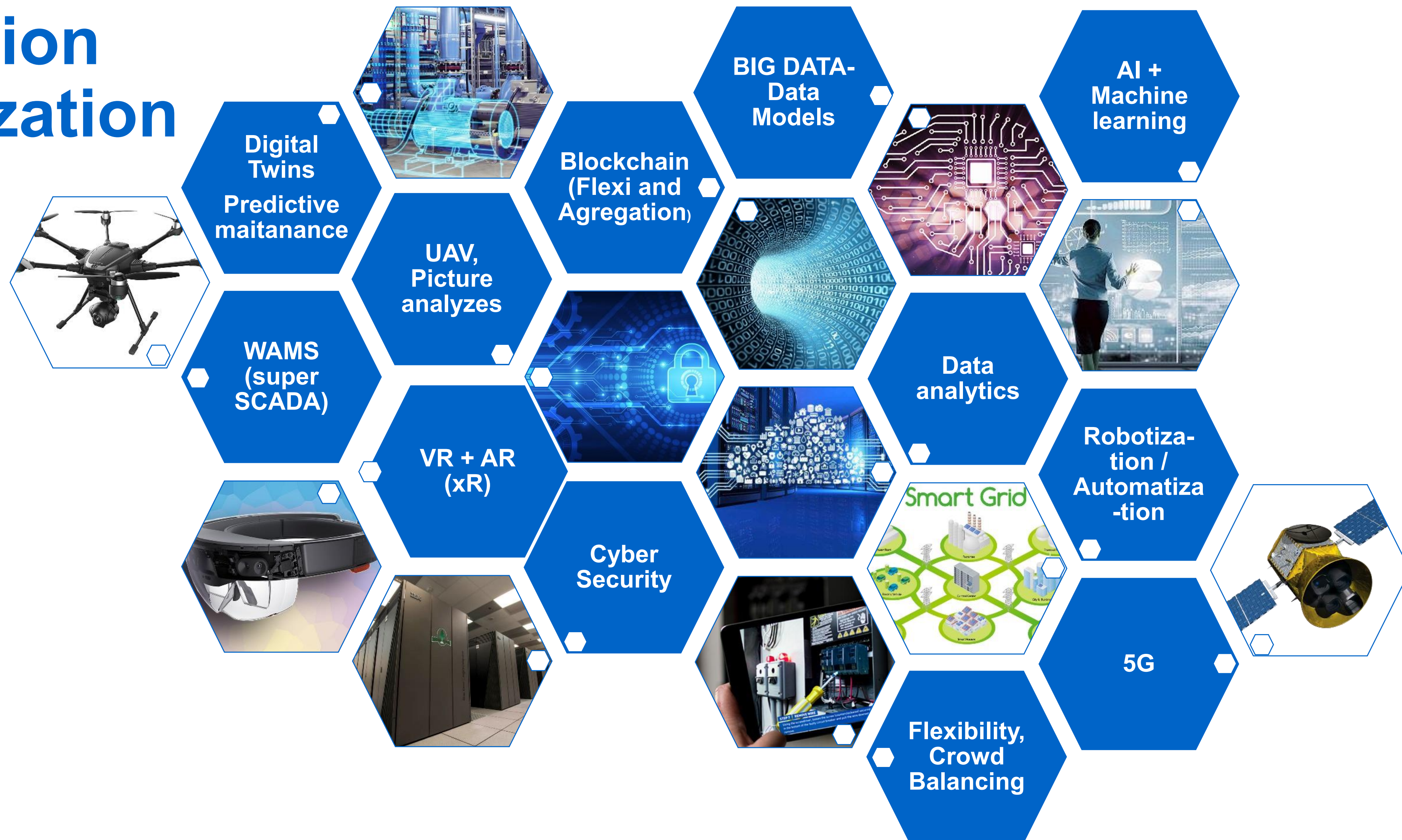
- We operate the **high-voltage power grid**, ensure **transmission** from generation to consumption & across the border.
- Power can't be stored; **production must = consumption** (grid **balancing**)
- **Security** and **stability** are the primary objectives.

But the energy transition is.... transformative!



- **Decommissioning** of large conventional **power plants** = loss of stability and balancing resources
- Increasing penetration of **intermittent (uncontrollable) renewables**
- Increasing **electrification** of all sectors

Innovation Digitalization AI



Grid Pulse Challenge Overview



What we want you to create?

An **AI-powered map-based view** of the power grid. You will turn complex data into clear operational insight.

Operational Relevance and Timeliness

The challenge prioritizes timely, clear, and actionable insights over theoretical optimization for grid reliability and risk mitigation.

Usable Tools and Prototypes

Participants are encouraged to develop practical dashboards, alert systems, or analytical tools that fit operator workflows.

Enhancing Operational Awareness

The guiding question is how to transform data into actionable awareness to improve grid stability and operational control.

Operational Context of the Transmission Grid



Real-Time Grid Balancing

Transmission grids balance electricity generation and consumption continuously to maintain system stability.

Increasing Grid Complexity

Renewable energy and decentralization increase variability and unpredictability in power flows.

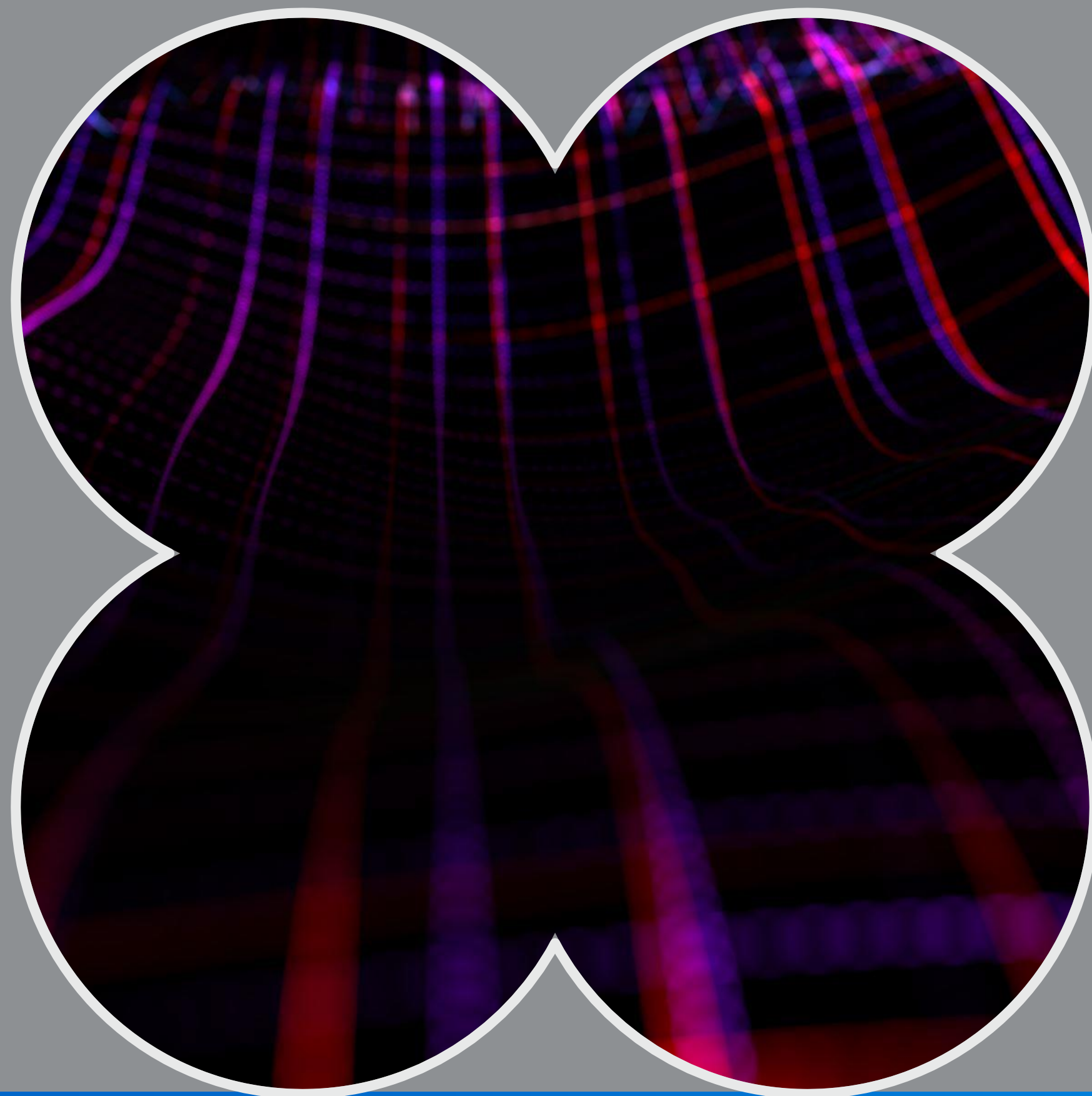
Operational Challenges

Operators face high cognitive loads processing diverse data under time pressure for quick decision-making.

Focus on Situational Awareness

The core challenge is converting abundant data into clear, actionable situational awareness for operators.

Grid Pulse as a Live System State



Grid Pulse Metaphor

Grid Pulse symbolizes a live, evolving state of the transmission system, similar to a medical pulse indicating health.

Real-Time Indicators

Includes key metrics like line loading, frequency deviations, congestion risks, and emerging anomalies.

Interpretability Over Data Volume

Focus on synthesizing data to highlight trends and risks for quick operator understanding, not raw numbers.

Time-Aware System Dynamics

Grid Pulse captures temporal changes and trajectories, representing the grid as a dynamic living system.

What Participants Are Expected to Build



Build a visual grid model

- Map-based representation of the grid
- Nodes = substations, generation, large consumption
- Focus on clarity and usability

Node data

- Static: capacity, type, limits
- Dynamic: current production/consumption, state, trends
- Clear visual differentiation

Predictive data integration

- Weather forecast
- Satellite / cloud cover data
- Connect external signals to grid behavior

From state to prediction

- Short-term outlook (minutes / hours)
- Highlight risks and expected changes

Dispatcher chatbot

- Natural language interface

Possible Technical and Conceptual Approaches



Data Analytics Techniques

Focus on descriptive and diagnostic analysis of historical and live data to reveal patterns and correlations.

Machine Learning and AI

Apply predictive models for forecasting overloads and detecting anomalies in system behavior.

Visualization Solutions

Develop intuitive dashboards to communicate system states clearly, minimizing operator cognitive load.

Scenario Simulation

Enable exploration of 'what-if' scenarios to understand impacts of interventions or events.

Overview of Available Data Types



Time-Series Data

Time-series data track power flows, voltages, and frequency at regular intervals for grid monitoring.

Topological Data

Topological data map grid structure including nodes, lines, and connections to contextualize measurements.

Forecast Data

Forecast data predict generation and consumption to anticipate risks and support planning.

Event Data

Event data capture outages, switching actions, and disturbances to explain sudden system changes.

... find what you can. **We will provide you with:**

Topological data, grid state data, load flows (CSV or JSON)

Key Technical and Design Expectations

- Clarity and visual strength
- Practical usefulness
- Connection of data and prediction
- Interpretability



Summary

- Make the grid **visible**
- Make it **understandable**
- Make it **predictable**

- Have **fun!**

(And win the **PRIZE** :-))





... ONE DAY, YOUR TOOL CAN BE HERE

AND IT WILL HELP KEEP THE LIGHTS ON

ENSURING BALANCED ENERGY FLOW

THANK YOU FOR YOUR ATTENTION

Jan, Premysl, Vit – reach out with any questions, we'll be around

